

AKTU Lucknow

REGULAR EXAMINATION ? 2026

Computer Organization and Architecture ? Regular Examination, 2026

Subject Code: CS304 Semester: 3 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Basic Structure of Computers)

1. List the advantages and limitations of Basic Structure of Computers. Explain any two in detail. [12 Marks]
2. Compare and contrast the key concepts involved in Basic Structure of Computers. [8 Marks]
3. Describe the working principle of Basic Structure of Computers with necessary equations. [8 Marks]

SECTION B (Unit 2: Machine Instructions and Addressing)

4. Compare and contrast the key concepts involved in Machine Instructions and Addressing. [12 Marks]
5. Derive the expression related to Machine Instructions and Addressing and explain each term. [12 Marks]
6. Describe the working principle of Machine Instructions and Addressing with necessary equations. [12 Marks]

SECTION C (Unit 3: ALU and Data Path Design)

7. Define ALU and Data Path Design. Explain its significance in Computer Organization and Architecture with a suitable example. [12 Marks]
8. Solve a problem related to ALU and Data Path Design. Show all the steps clearly. [12 Marks]
9. Discuss the engineering applications of ALU and Data Path Design. [12 Marks]

SECTION D (Unit 4: Memory Hierarchy and Cache)

10. Discuss the engineering applications of Memory Hierarchy and Cache. [12 Marks]
11. Compare and contrast the key concepts involved in Memory Hierarchy and Cache. [12 Marks]
12. List the advantages and limitations of Memory Hierarchy and Cache. Explain any two in detail. [12 Marks]

SECTION E (Unit 5: Input/Output Organization)

13. Define Input/Output Organization. Explain its significance in Computer Organization and Architecture with a suitable example. [12 Marks]

14. Discuss the engineering applications of Input/Output Organization. [8 Marks]

15. How is Input/Output Organization implemented? Give an algorithm / procedure and explain. [8 Marks]

*** End of Question Paper ***